Sahanka AI ee Dayax-gacmeedka: wireframes & user flow

Warbixin 12 bog ah oo ku saabsan sida loo habeeyo muuqaalka iyo socodka feature-ka cusub

FEATURE CUSUB · AI SATELLITE SURVEY RECOGNITION

Sawir dayax-gacmeed → waddooyin → blocks → plots → soo saar

Prompt-ka aad soo gelisay wuxuu codsanayaa in appka jira lagu daro aqoonsiga otomaatiga ah ee waddooyinka sahanka iyo blocks-ka, **adigoo aan dib u dhisin appka** oo aan beddelin xisaabta qaybinta. Warbixintani waxay qeexaysaa **sida muuqaalka iyo socodka loo habeeyo**.

Cusub — raadin otomaatig ah, ma jiro sawir la soo geliyo: appku wuxuu **isagu raadiyaa** sawirrada dayax-gacmeed ee xuduudka beerta, laga bilaabo sanadka hadda (2026) **ilaa 10 sano dib** (2016). Wuxuu dooranayaa sawirka sahanka ugu fiican ee muuqda — **calaamadaha tractor ama cagaf ee labada dhinac waddada** — wuxuuna hubinayaa in labada xariiq ay u dhaxeeyaan **15 m** iyo in block-gu yahay **90 × 48 m**.

10	21	15 m	0
sano oo dib loo raadiyo	shaashadood oo wireframe ah	waddo · block 90×48	ogolaansho otomaatig ah

Waxa prompt-ka laga fahmay

- **Gelinta:** GeoTIFF, PNG/JPG oo georeferenced ah, ama screenshot oo la georeference-gareeyo dhibco kontorool (control points) — **weligeed pixel looma tixgelin goob dhab ah**.
- **Aqoonsi:** waddooyin, blocks, centre lines, ballaca waddada, isgoysyada, plots (haddii ay muuqdaan) iyo bearing.
- **Dib-u-eegis dad:** wax kasta oo AI ogaado waa la eegayaa, la hagaajinayaa, la ogolaanayaa ama la diidayaa.
- **Xeerarka jira ee aan la taaban:** block 90 × 48 m · waddo 15 m · plot buuxa 18 × 24 m · half-plots (18×12 iyo 9×24) · 10 plots/block.
- **Offline & gaarnimo:** inference gudaha taleefanka; sawirka dhulka gaarka ah looma soo bandhigo adeeg dibadeed.
- **Hawl-gal:** module gooni ah oo ku xirma khariidadda, lakabyada, bearing engine-ka iyo subdivision engine-ka jira.

Lix mabda’ oo muuqaalka hagaya

1 · Dad ayaa go’aan gaara
AI wuxuu soo jeediyaa; isticmaaluhu wuu ogolaadaa.

2 · Shaki muuqda
Aan la hubin = jaalle xariijin goos-goos ah.

3 · Goobta dhabta ah
Georeference kahor baaritaanka; RMSE la muujiyo.

4 · Xeerarka la qufulay
90×48, 15m, 18×24 — akhris keliya.

5 · Offline & gaar
Banner: “Sawirkaaga ma baxayo”.

6 · Dhig la ogaaday vs la sameeyay
Plots-ka engine-ka ha ahaado midab kale.

Ogeysiis: waxaan qoray keliya muuqaalka, wireframes-ka iyo user flow-ga. Koodhka appka lama arkin, sidaas darteed lama heli karo go’aan teknikeed oo ku saabsan moodelka AI-ga ama halka koodhka lagu daro — waxaa taas hore loo sahamin doonaa sida prompt-ku farayo (“inspect existing architecture first”).

Prompt-ka → shaashadaha: khariidad buuxda

Prompt-ka, feature-ka raadinta otomaatiga ah iyo halka muuqaalka ay ku muuqanayaan

#	Qaybta prompt-ka	Waxa muuqaalka ku jira	Bogga
1	Raadin otomaatig ah · 10 sano	Setup (xuduud → bbox, isha, sanadaha) · scan · liiska natiijooyinka	P5
2	Hubinta 15 m & 90×48	Calaamadaha labada dhinac · cabbir · tolerance · warbixin	P6
3	Gelinta gacanta (kale)	Soo geli sawir + control points (haddii raadinta wax la waayo)	P7
4	Ogaanshaha waddooyinka	Baaritaan + overlay waddooyin (cyan / jaalle)	P8 P9
5	Aqoonsiga blocks	Overlay blocks leh ID + kalsooni %	P9 P10
6	Vector-ka waddooyinka	Dhex-xariiq ≠ Ballac waddo (labo style)	P9 P11
7	Aqoonsiga plots	“La ogaaday” vs “Engine ayaa sameeyey”	P11
8	Bearing	Dial + manual + reference-edge + preview	P8
9	Dib-u-eegis & hagaajin	Ku dar / tirtir / edit / isku-dar / kala-jar / ogolow / diid	P9 P10
10	Gudbinta engine-ka	Shaashad u gudbin + xeerar la qufulay	P11
11	Soo saarka GIS	Roads · Blocks · Plots · Combined · GPKG + CRS	P11
12	Offline, gaarnimo & internet	Moodel offline; ogolaansho cad ee raadinta sawirka (internet)	P5 P8
13	Ansixinta & tijaabada	Precision · recall · false · positional	P11 P12
14	Xeerarka adag	Guard rails: aan la ogolaan karin otomaatig, aan la beddeli karin	P12

Waxa shuruudaha prompt-ku u beddelayaan UX

“Weligeed pixel ha u tixgelin goob”

→ Baaritaanka waa **la xidhi** ilaa georeference la xaqiijiyo; badhanka “Baar” wuu dami yahay oo sababta ayaa lagu qoraa.

“Ha ogolaan wax shaki ah otomaatig”

→ Dhammaan waxa ay AI ogaato waxay bilaabmaan **“Shaki”**; midna ma xaqiijmo ilaa isticmaaluhu riixo “Ogolow”.

“Ha beddelin xisaabta engine-ka”

→ Shaashadda gudbinta waxay muujisaa dhererka iyo ballaca jira **akhris keliya ah**; beddelid waxaa laga sameeyaa Settings-ka jira.

“Ha u soo bandhigin plots aan muuqan inay la ogaaday”

→ Plots-ka engine-ku sameeyo waxay leeyihiin tag **“Engine”** iyo xariiq kale; kuwa la arkay waa **“La ogaaday”**.

“Ha dhihin survey-grade”

→ Warbixinta waxay ku jirtaa digniin joogto ah: “Natiijadu waa kaalmo muuqaal ah, ma aha cabbir sahan ah.”

“Ha u soo bandhigin AI dibadda”

→ Baaritaanku waa offline. Raadinta sawirka waxay u baahan tahay internet: provider-ku wuxuu arkaa **aagga (bbox)** la weydiistay — ogolaansho cad ayaa lagu weydiiyaa; xuduudka polygon-ka lama diro.

User Flow: socodka buuxa ee isticmaalaha

Laga bilaabo khariidadda ilaa soo saarka — go’aamada iyo laamaha

A · Helista sawirka ugu fiican (otomaatig — ma jiro gelin gacanta)

Khariidad
Riix “+”

→ **Sahan AI**
Dooro “Sahan AI”

→ **1 · Sawir**
Dooro habka

→ **◇ Habka?**
Auto (caadi) · Gacanta (kale)

Raadin otomaatig ah
Xuduudka beerta → bbox · ogolaansho internet

→ **Sawirrada 2026 → 2016**
Metadata: taariikh · cabbir pixel · daruur · daboolid

→ **Qiimee muuqaalka**
Raadi calaamadaha tractor/cagaf ee labada dhinac waddada

◇ Cabbirku ma saxan yahay?
Waddo 15 m · block 90×48 (tolerance: Settings)

→ **HA → Dooro sawirka**
Top 5 · isbarbar-dhig sanado · goob dhab ah toos

→ **MAYA → sanad kale**
Sii raadi ilaa 2016; haddii la waayo → gacanta

→ **Gacanta (kale)**
Soo geli sawir + control points · P7

↓

B · Baaritaan

3 · Dooro
Waddooyin · Blocks · Plots; moodel offline

→ **Bearing**
Auto ▸ manual ▸ reference-edge

→ **4 · Baar**
Gudaha taleefanka · progress · jooji

→ **Natiijo → “Shaki”**
Dhammaan waxay bilaabmaan aan la xaqiijin

↓

C · Dib-u-eegis (wareegga ugu muhiimsan)

5a · Waddooyin
Ogolow · Diid · Edit · Ku dar

→ **5b · Blocks**
Ogolow · Diid · Corners · Isku-dar · Kala-jar

→ **◇ Wax shaki ah ma hadhay?**
Haa → ku noqo · Maya → sii wad

↓

D · Gudbin & soo saar

6 · U gudbi
Kaliya blocks la xaqiijiyey

→ **Engine-ka jira**
90×48 · 15m · 18×24 · half-plots · ID

→ **Plots**
Engine = midab 1; la ogaaday = midab 2

→ **7 · Soo saar**
Roads · Blocks · Plots · Combined · GPKG

Gelitaanka

Laba meelood ayaa la gaari karaa: (a) “+” ee khariidadda, (b) tab-ka “Khariidad” ee kiis jira (“Sahan AI” rail ku jira).

Laamaha khaladka

Moodel ma jiro → soo deji moodel · Sawir weyn → dhig yaraan · Baaritaan fashilmay → isku day / jooji · Aan CRS lahayn → control points.

Dib u laabasho

Tallaabo kasta wuu dib loo laaban karaa; geometry-ga la hagaajiyey waa la xafidaa (draft) ilaa la xaqiijiyo.

Halka feature-ku ku fadhiyo & xaaladaha geometry

Gelitaan, wizard-ka 5 tallaabo, iyo state machine-ka natiijooyinka



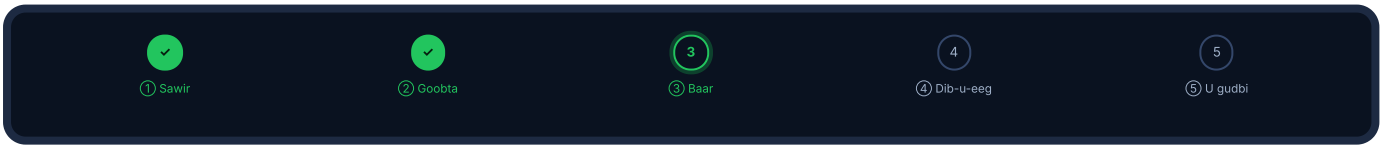
1 Menu-ga “+” — ficil cusub, mid ka mid ah

Meesha feature-ku yaal

- **Ma jiro tab hoose cusub.** Appka wuxuu hayaa 4 tab; Sahan AI waa **wizard buuxa** oo ka furma “+” ama rail-ka khariidadda.
- Marka la bilaabo, bar-ka hoose wuu qarsoomaa; waxaa muuqda stepper 5 tallaabo (Sawir · Goobta · Baar · Dib-u-eeg · U gudbi).
- Marka la dhammeeyo, natiijadu waxay noqotaa **lakabyo caadi ah** (Roads · Blocks · Plots) oo ku jira “Lakabyada” khariidadda.
- Draft-ka waa la keydiyaa: haddii appka la xiro, dib ayaa loo furi karaa.

Sababta: mid ka mid ah “+” wuxuu isticmaalaha u ogeysiiyaa in tani ay tahay *habka kale ee abuurista nashqad*, halka subdivision engine-ku sii jiro sida hore.

Stepper-ka wizard-ka



Xaaladaha geometry (state machine)

La ogaaday
AI ayaa ogaaday · **Shaki**

→

Ogolow
→ **La xaqiijiyey**

Diid
→ **La diiday**

Edit
→ **Wali shaki** ilaa la ogolaado

Ku dar gacanta
→ **Gacanta · la xaqiijiyey**

→

Isku-dar / Kala-jar
→ Cusub = **Shaki** ilaa la ogolaado

→

U gudbi
Kaliya **La xaqiijiyey** iyo **Gacanta**

Dhammaan isbeddelka waxaa la qoraa **log** (waa la rogi karaa — Undo).

Wireframes A: Raadin otomaatig ah (10 sano dib)

Tallaabo 1 · Xuduudka beerta → sawirrada 2026→2016 → liiska la qiimeeyey



1 Setup — isha, sanadaha, ogolaansho



2 Scan — sanad-sanad, qiimaha muuqaalka



3 Natiijo — sawirrada ugu fiican

Sida raadintu u socoto

1. Isticmaaluhu **ma gelinayo sawir** — xuduudka beerta ee jira ayaa isu beddelaya bbox (aag raadin).
2. Sanadaha: laga bilaabo sanadka hadda ilaa **10 sano dib**; wuxuu ka bilaabmaa kan ugu cusub oo sii socdaa.
3. Sanad kasta wuxuu marka hore dhaafaa shaandhayn degdeg ah: **cabbir pixel, daruur, daboolka aagga** — kuwa aan fiicnayn waa laga boodaa.
4. Kuwa dhaafa waxaa la qiimeeyaa: **muuqaalka sahanka** (calaamadaha tractor/cagaf) iyo cabbirka (bogga 6).

Ogaysiiska internet

Raadinta waxay u baahan tahay internet, sidaas darteed waxaa la muujiyaa ogolaansho cad: provider-ku wuxuu arki karaa aagga laga codsaday. Marka sawirka la doorto, baaritaanka waa **offline**.

Goob dhab ah toos

Sawirrada tile-ka waa georeferenced, sidaas darteed **control points looma baahna** — kaliya habka gacanta (P7).

Wireframes B: Hubinta waddada 15 m & block 90 × 48

Cabbirka labada xariiq (tractor/cagaf) iyo cabbirka block-ka — qiimayaasha waa tusaale



1 Sawirka la doortay — calaamadaha sahanka



2 Waddo: dhexda labada xariiq = 15 m



3 Block: 90×48 · farqi · 105×63

Sida hubinta u socoto

1 **Waddo:** la raadiyo labada xariiq ee isku dhow (calaamadaha tractor/cagaf); la cabbiro dhexda u dhexeysa → waa inay u dhawaato **15 m**.

2 **Block:** dhererka iyo ballaca block kasta → **90 × 48 m**; dhexda-dhexda waddooyinka ≈ 105 × 63 m (haddii block-ku u dhexeeyo waddooyin 15 m — soo jeedin).

3 **Tolerance:** lagu qeexo Settings (tusaale max(1.0 m, 2×pixel)) — AI ma beddeli karo.

4 **Natiijo:** ✓ ku jira · ⚠ dib-u-eeg · × ma dhaafin; midna otomaatig looma ogolaado.

Cabbir ≠ sax

Qalad cabbirka pixel ayaa jiri kara: pixel 0.5 m ayaa keeni kara ±0.5 m. Sidaas darteed tolerance-ku waa inuu ka weyn yahay pixel-ka, warbixintuna waa inay muujisaa. Tani waa hubin muuqaal ah, ma aha cabbir sahan ah (survey-grade).

Haddii cabbirku khaldamo

Dooro sanad kale (isbarbar-dhig) ama sii wad dib-u-eegis: block-yada ⚠ waxay noqdaan “Shaki” oo gacanta ayaa loo hagaajiyaa.

Wireframes C: Habka kale — soo geli gacanta & georeference

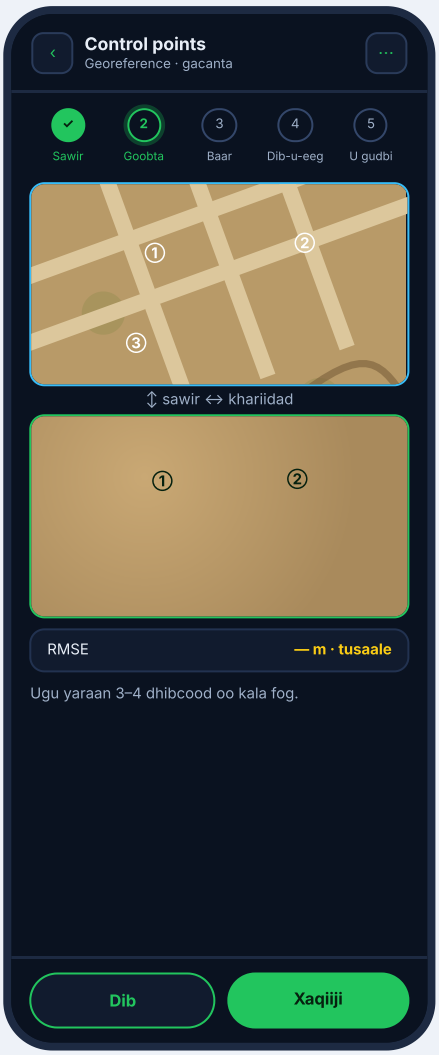
Tallaabo 1–2 · Waxaa la xidhaa baaritaanka ilaa goobta dhabta ah la hubiyo



1 Soo geli — 3 nooc + banner gaarnimo



2 Hubi goobta — CRS / extent



3 Control points — sawir iyo khariidad

Xeerarka muuqaalka

1 Banner “🔒 Sawirka ma baxayo” wuu joogaa marka la gelayo — ka hortag aamin-la’aan.

2 Haddii sawirku CRS leeyahay: kaar cagaar “Georeferenced”; haddii kale: kaar jaalle ah + badhan “Bilow georeference”.

3 “Baar” waa dami ilaa georeference la hubiyo — sababta ayaa lagu qoraa hoosta.

4 Control points: sawir kor, khariidad hoos, isla tirooyin (①②③); RMSE si toos ah ayaa loo tiriyaa.

Qiimaha RMSE

Tirada “— m” waa meel-buuxiye. Heerka aqbalaadda (tusaale: ≤ 2× cabbirka pixel) waa in lagu qeexaa Settings, ma aha mid AI go’aamiyo.

Wireframes D: Dooro · Baar · Bearing

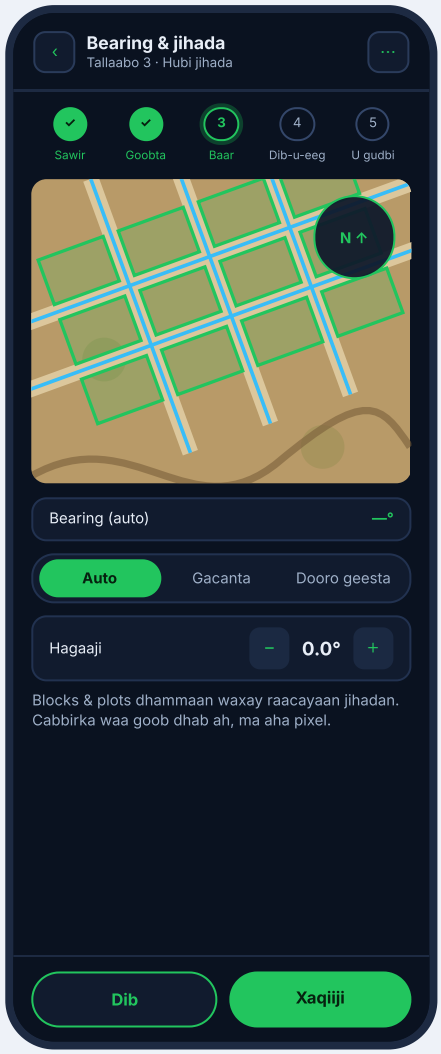
Tallaabo 3 · Offline, gaarnimo iyo jihada nidaamka



1 Dooro baaritaanka & moodelka offline



2 Progress — waddooyinka ugu horreeya



3 Bearing: auto / gacanta / geesta

Qodobbada muhiimka

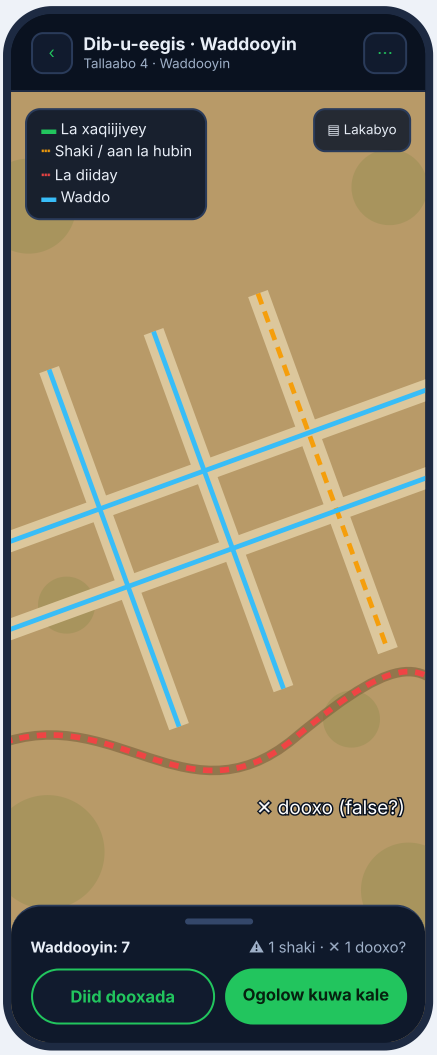
1. Moodel-ka waa la maareeyaa (soo dejis, tirtir, cabbir) — ma jiro soo-dejis qarsoodi ah.
2. Baaritaanka way joogsan karaa; natiijooyinka qayb ahaan waa la hayaa.
3. Bearing waxaa laga xisaabiyaa goobta (CRS), ma aha pixel; preview-gu wuxuu muujiyaa jihada ka hor inta aan la xaqiijin.
4. “Dooro geesta” (reference-edge): isticmaaluhu wuxuu taabtaa xariiq si uu jihada ugu qeexo.

Android

Sawirka weyn waa la jajabiyaa (tiles) si xusuusta u badbaado; hawsha waxay ku socotaa background service oo leh notification. Tijaabada moodelka lama sameyn — waa in la sahmiyo.

Wireframes E: Dib-u-eegis (shaashadda ugu muhiimsan)

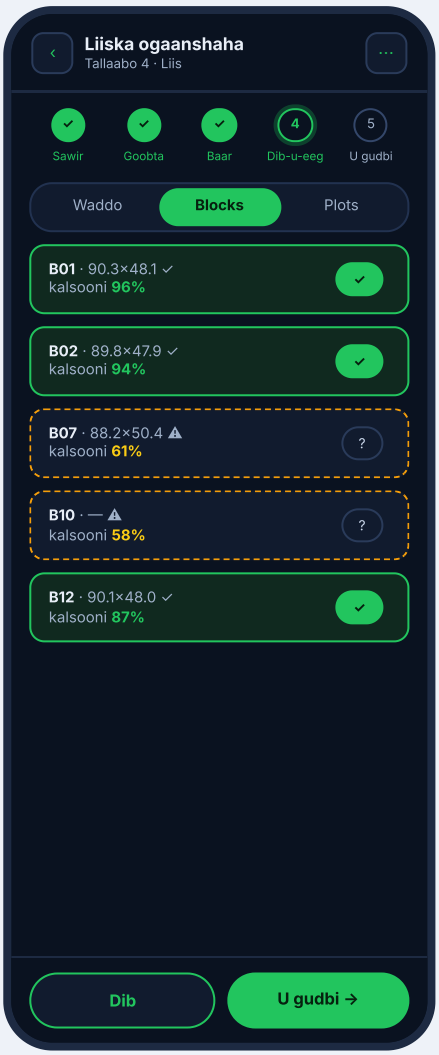
Tallaabo 4 · Sawirka asalka + overlay + kalsooni + shaki



1 Waddooyin — cyan; jaalle = shaki; cas = dooxo



2 Blocks — ID + kalsooni % ; jaalle goos-goos = shaki



3 Liiska — cabbir (tusaale) + shaki/xaqiijin

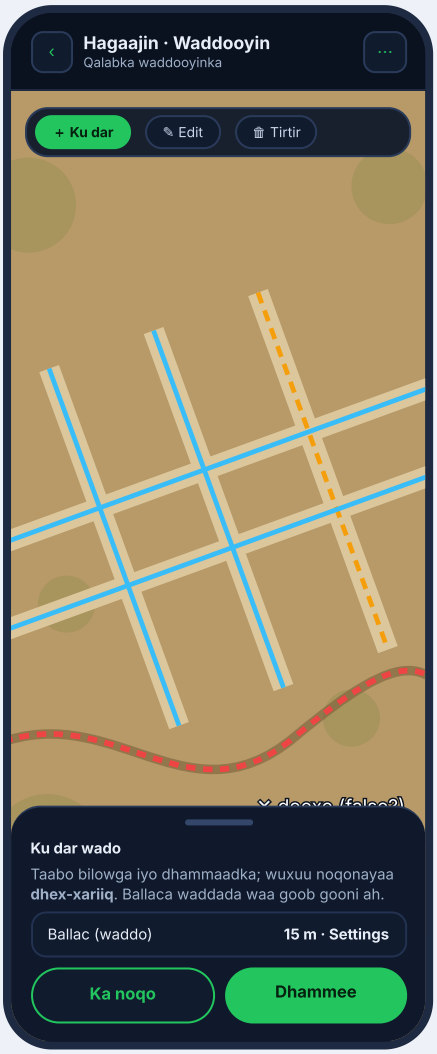
Qaabka muuqaalka

Xaalad	Xariiq	Midab
La xaqiijiyey	buuxa	Cagaar
Shaki / aan la hubin	goos-goos	Jaalle-oranji
La diiday	goos-goos, dabacsan	Cas
Gacanta lagu daray	buuxa	Buluug

Muhiim: jaalle = “aan la hubin” waa in uu ahaadaa **xariiq goos-goos** oo ay la socoto calaamad ?, si ay u muuqato xitaa haddii midabka uu aan la kala saari karin. Sawirka asalka waa la arki karaa (slider “Asalka ↔ Overlay”). Tirooyinka sawirka waa tusaale.

Wireframes F: Qalabka hagaajinta

Ku dar · tirtir · edit · geesaha · isku-dar · kala-jar · ogolow/diid



1 Waddooyin — ku dar / edit / tirtir



2 Block — geesaha (corners) & ogolow/diid



3 Isku-dar & kala-jar

Habdhaqanka

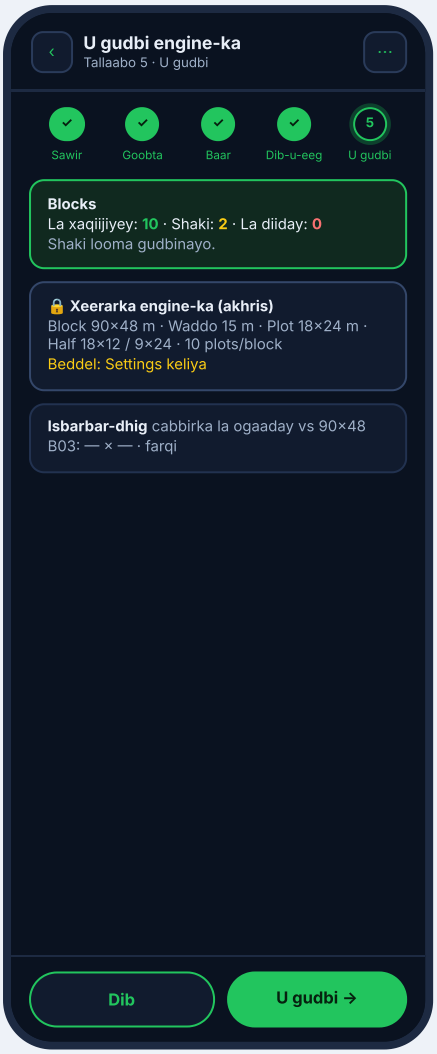
1. Toolbar-ku wuxuu ku jiraa dusha khariidadda (chips), sheet-ka hoose wuxuu muujiyaa faahfaahinta.
2. Geesaha waxay leeyihiin **handle weyn** (≥ 48dp touch); tilmaan magnet ah marka la dhawaado waddo.
3. Block cusub oo ka dhashay isku-dar/kala-jar waa **Shaki** — ma ogolaado otomaatig.
4. Undo / Redo joogto ah ayaa ku jira header-ka.

Ballaca waddada ≠ dhex-xariiqda

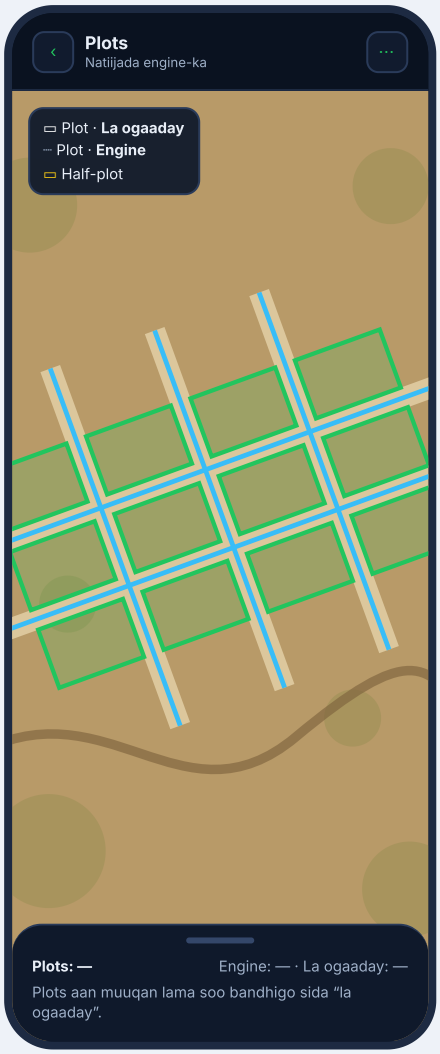
Prompt-ku wuxuu digayaa in aan isku khaldin dhex-xariiqda iyo xuduudka reservation-ka. Sidaa darteed waxaa la muujiyaa **labo style**: xariiq khafiif ah (centre line) iyo polygon dabacsan (ballac) — kan dambe kaliya marka ballaca la qiyaasi karo.

Wireframes G: U gudbi · Plots · Soo saar

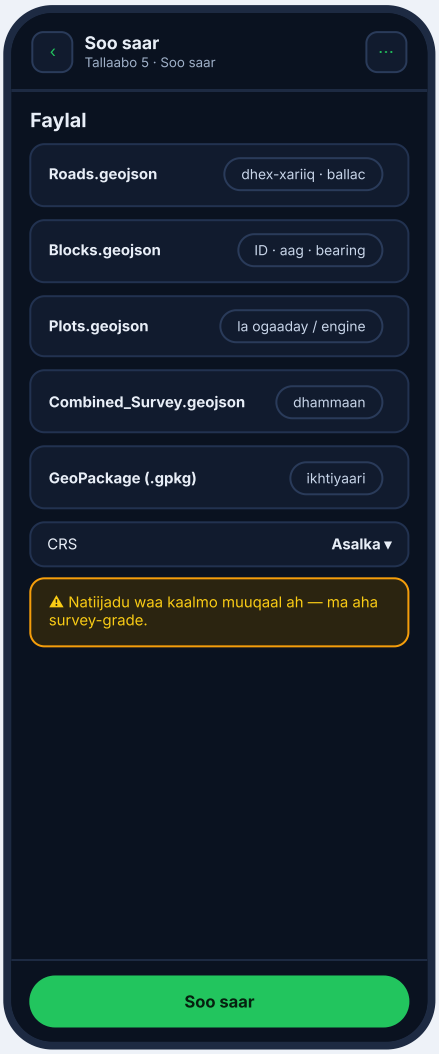
Tallaabo 5 · Xeerarka la qufulay, plots la ogaaday vs engine, iyo CRS



1 U gudbi — kaliya la xaqiijiyey



2 Plots: la ogaaday vs engine



3 Soo saar — 4 fayl + GPKG + CRS

Xeerarka

1. "U gudbi" wuu dami yahay haddii 0 block la xaqiijiyey; wuxuu u gudbiyaa kaliya kuwa la xaqiijiyey.
2. Xeerarka 90×48, 15m, 18×24 iyo half-plots waa **akhris keliya** — AI ma beddeli karo.
3. Plots-ka engine-ku sameeyo waxay leeyihiin tag + xariiq gooni ah.
4. Soo saarka waxaa ku jira CRS asalka iyo xulasho beddelid; digniin survey-grade joogto ah.

Warbixinta ansixinta (tab ku jira "Dib-u-eeg")

Cabbirka	Qiimaha
Precision	— (tusaale)
Recall	—
Kuwa been ah (false)	—
Qalad goob (m)	—

Afarta cabbirood gooni gooni ayaa loo muujiyaa, sida prompt-ku farayo.

Qeexitaanka muuqaalka, xaaladaha & hawlgal

Visual language · kalsooni · khaladaad · Android · tallaabooyin

Luqadda muuqaalka

La xaqiijiyey
xariiq buuxa · cagaar

Shaki
goos-goos · jaalle + “?”

La diiday
goos-goos · cas · dabacsan

Gacanta lagu daray
buuxa · buluug

Plot “Engine”
khafiif · cawl · tag “Engine”

Heerarka kalsooni (soo jeedin)

Heer	Muuqaal	Ficil
≥ 85%	Sare	Waa “Shaki” ilaa la ogolaado
60–84%	Dhexe	Dib-u-eeg lagama maarmaan
< 60%	Hoose	Calaamad ⚠ · gacanta

Heerarka waa tusaale; waa in lagu hagaajiyo tijaabo. “Ogolow kuwa sare” waa badhan isticmaale riixo — marna ma dhacayo otomaatig.

Xaaladaha khaladka (loo qorsheeyo)

- Sawir aan CRS lahayn → georeference.
- RMSE sare → digniin + hagaaji.
- Waddooyin aan muuqan → ku dar gacanta.
- Xusuusta oo yar → tiles / yaree.
- Baaritaan fashilmay → natiijo qayb ah la hayaa.

Android & offline

- Inference gudaha taleefanka (ONNX Runtime Mobile / TFLite — la tijaabiyo).
- Draft la keydiyo; shaashadda ha xirmin.
- Moodel la maareeyo (cabbir, tirtir).
- Wax uploads ah ma jiro iyada oo aan ogolaansho jirin.

Isha sawirrada & xaddidaadaha (raadinta otomaatiga ah)

Isha	Waxa ay bixiso	Ogeysiis
Esri World Imagery Wayback	Release-yo taariikhi ah; metadata goob walba: **taariikh, isha, provider, cabbir pixel, saxnaan** (sida ku qoran wayback-core).	Waa in lagu hubiyo shuruudaha Esri ka hor soo-dejin/baaritaan otomaatig ah.
Sentinel-2 / Landsat	Bilaash; pixel ≈ 10 m / 30 m.	Waddo 15 m ≈ 1.5 pixel → calaamadaha tractor ma muuqdaan; **ma habboona** hubinta 15 m.
Commercial (Maxar, Airbus, Planet)	Sub-meter; API leh furaha adiga.	Lacag + shuruud; helitaanka aagga lama xaqiijin.
Sawirkaaga (drone / faylka)	Habka gacanta (P7).	Had iyo jeer waa la heli karaa.

Cabbirka pixel ee loo baahan yahay (qiyaas): ≈ 0.5 m ama ka yar si waddo 15 m ay u noqoto ≥ 30 pixel; tolerance = max(1.0 m, 2 × pixel) — waa soo jeedin, lagu hagaajiyo tijaabo. Haddii aagga uu heli waayo sawir wanaagsan, appku wuxuu sheegayaa — ma sameynayo natiijo been ah.

Jadwalka hawlgal

#	Tallaabo	Natiijada
0	**Raadinta:** Wayback metadata (taariikh/cabbir) → shaandhayn → qiimeynta calaamadaha → cabbirka UTM 38N.	Liis sawirro la qiimeeyey
1	**Sahan koodh:** khariidad, layers, GeoJSON, bearing & subdivision engine — meelaha ku-xidhidda.	Warqad ku-xidhid
2	Georeference module (control points, RMSE, CRS).	Sawir → goob dhab
3	PoC moodel offline oo yar (waddooyin) + tijaabo cidla’.	Precision / recall
4	Review UI (overlay, shaki, liis) + qalabka hagaajinta.	Wizard 1–4
5	U gudbinta engine-ka + soo saarka (Roads/Blocks/Plots/Combined/GPKG).	Wizard 5
6	Tijaabada: xallin, iftiin, saxaraha, dooxo, geoloc/ma-geoloc, qalad goob.	Warbixin ansixin

Xusuusin: waxa muhiimka ah ma aha inuu AI saxan yahay, ee inuu isticmaaluhu arki karo, hagaajin karo oo go’aamin karo ka hor inta aan engine-ka la gelin.